Preflight: making an SO-101 policy generalise to new scenes, objects and tasks

Experiments, funnels and verdicts (simulation only)

Preflight, Nebius × NVIDIA Physical AI hackathon

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Abstract

We fine-tune GR00T N1.7 for pick-and-place on a MuJoCo twin of the SO-101 arm and try seven ways to make it generalise. Every candidate is scored on the same six paired suites (96 rehearsals each) against the v5 baseline, with scene-by-scene McNemar tests and a promotion gate. DAgger with upweighted expert recoveries is the one lever that passes the gate, twice: round 2 (pooled +26 of 576 over v5) and round 3 (+19 more, now served). A fairly trained FLUX 3 Action LoRA loses on every suite despite predicting the right actions open-loop. A third DAgger round is the first candidate to move new objects (61 vs 50 of 96, $p = 0.043$), though its pooled gain over round 2 is within noise. All results are in simulation; the physical arm has not been delivered.

1 Setup

Policy. `hqfang/groot-so100_101` (GR00T N1.7, pretrained on 1,209 SO-100/101 datasets), fine-tuned with LeRobot 0.6.1. **v5** adds 800 scripted twin demos (12k steps, batch 8, lr 2.5×10^{-5}) and is the baseline for every comparison.

Benchmark. Six suites, each 12 scenes × 8 perturbed rehearsals = 96, seed 42: `held_out_objects` (objects never in training), `seen_vocabulary`, `clean_many`, `clean_top_down_many`, `real_1_pomade_tin_many`, `in_path_pomade_tin_many`. Re-running one checkpoint moves a suite by up to ±9, so differences are judged only by paired McNemar tests on the scenes where two checkpoints disagree. A candidate is *promoted* only with a pooled paired gain and no suite significantly worse; the gate was fixed before any round was scored. External check: so101-benchmark `stack_blue_on_red`, 50 episodes.

Failure labels. Each failed rehearsal is labelled by `FailureWatch`: *no_contact* (pick block never moved), *pushed*, *dropped*, *fell_off*, *holding* (still lifted at timeout), *knocked_place*.

2 Experiments

2.1 DAgger with upweighted recoveries (the one that works)

Each round rolls the current policy out on 200 fresh training scenes × 4 variants, snapshots every failure (simulator state plus scene), keeps the most uncertain within a recording budget balanced across failure types, and lets the scripted expert record a recovery from each kept state. Table 1 gives the funnel with every denominator; failed recoveries (the expert could not stack from the state) are counted, not dropped.

Round (policy)	Rollouts	Succeeded	Failures kept	Recovered	Recoveries in training
1 (v5)	800	456 (57%)	207 of 330 recoverable	85 (41%)	85 of 2,485 (3.5%)
2 (v5)	800	494 (62%)	189 of 296 recoverable	72 (38%)	157 ×8 of 3,656 (≈ 34%)
3 (round 2)	800	574 (72%)	146 of 197 recoverable	59 (40%)	216 ×8 of 4,128
4 (round 3, RL expert)	800	614 (77%)	120 of 165 recoverable	88 (73%)	304 ×8 of 4,832
5 (round 3, RL ×2)	800	610 (76%)	113 of 160 recoverable	72 (64%)	216 ×8 + 160 ×2 of 4,448

Table 1: DAgger funnels. "Recoverable" excludes *holding*, which the open-loop expert cannot recover from a lifted state. Round 3 harvests with round 2; round 4 with the promoted round-3 policy, and its recoveries are recorded by the residual RL expert.

Suite (of 96)	v5	Round 1	Round 2	Round 2 vs v5
held_out_objects	52	47	50	$+12/-14$, $p = 0.85$
seen_vocabulary	75	76	76	$+8/-7$, $p = 1.0$
clean_many	81	82	86	$+7/-2$, $p = 0.18$
clean_top_down_many	83	77	85	$+6/-4$, $p = 0.75$
real_1_pomade_tin_many	74	78	**86**	$\mathbf{+14/-2}$, $\mathbf{p = 0.004}$
in_path_pomade_tin_many	73	76	81	$+13/-5$, $p = 0.10$
pooled (of 576)	438	436	**464**	+26: **promoted**
Round 3 (vs round 2)		481; held_out 61 vs 50, $+18/-7$, $p = 0.043$		+17: kept
Round 3 retrained		**483**; held_out 57 vs 50, $p = 0.30$		+19: **promoted**
Round 4 (vs round 3)		471; held_out 49 vs 57, $p = 0.17$		−12: kept
Round 5 (vs round 3)		455; held_out 41 vs 57, $p = 0.004$		−28: discarded

Table 2: Paired results. Round 1 (recoveries 3.5% of training) was kept, not promoted. Round 2 upweighted them to about a third and was promoted. Round-2 failures fell from 138 to 112 (*no_contact* 57 → 33) while *dropped* and *holding* did not fall.

2.2 FLUX 3 Action, trained fairly

A LoRA (rank 32) on `black-forest-labs/flux-3-action-so101`, trained on exactly v5's 800 demos (bridged into FLUX's joint units by a per-joint quantile map) for exactly v5's volume (96k samples in 12k updates of 8), never initialised from v5. It scores 50 of 576 against v5's 438, worse on every suite ($p < 0.001$ each), and 0 of 7 on so101-benchmark (stopped early: all timeouts). 358 of its 526 failures are *no_contact*. Executing 16 actions per chunk instead of 32 does not help (held_out 4 vs 3, clean_many 2 vs 9, $p = 0.039$ worse).

To separate the model from the data bridge we score its predicted chunk open-loop on 60 held-out demo frames, each fed the 7 frames before it (FLUX conditions on 8). Table 3 and Figure 1: the LoRA predicts the demo's actions three times better than holding the pose at 32 steps ahead; zero-shot FLUX is no better than standing still. The bridge sits inside this loop both ways. The closed-loop failure is therefore execution, not the mapping.

Steps ahead	Fair LoRA	Zero-shot FLUX	Hold pose
1	0.23	0.37	0.53
8	1.08	2.26	2.47
16	1.75	4.34	4.66
32	**2.71**	8.19	8.24

Table 3: Open-loop mean |action error| (LeRobot joint units) on held-out demo frames.

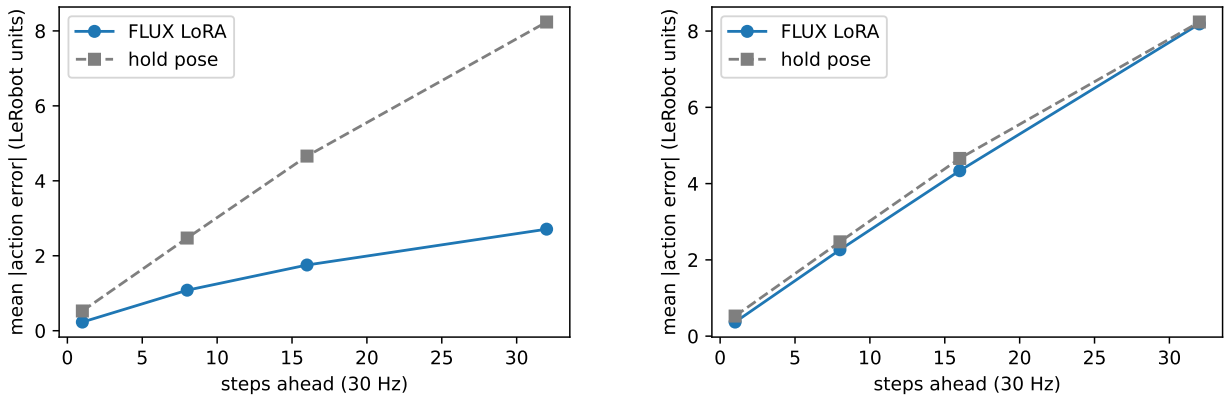


Figure 1: Open-loop error against horizon: fair LoRA (left) and zero-shot FLUX (right), each beside holding the pose.

2.3 Other experiments

- **Failure labels** (576 v5 rehearsals): no_contact 41%, pushed 25%, dropped 17% of failures.
- **Targeted demos (v6)**: recoveries after staged mistakes plus lookalike decoys; within noise of v5 on every suite.
- **External benchmark gap**: v5 scores 0/50 on so101-benchmark; the joint mapping and front camera match, the wrist-camera mount and table texture do not.
- **Visual domain randomisation (v7)**: the Seeed wrist mount alone drops v5 from 81 to 15 on clean_many's scenes; v7 lifts that to 53 but regresses clean_many itself to 58, and scores 1/50 externally. Not promoted.
- **Real-data co-training (v9)**: 400 ArmnetBench SO-101 teleop demos of new objects at 23% of samples score 373 vs 438; held_out_objects 48 vs 52 ($p = 0.59$), worse on clean_many, clean_top_down_many and real_1_pomade_tin ($p \leq 0.043$), 0/50 externally. Through the twin's kinematics 99.7% of the teleop episodes put the gripper under the table; fitted joint offsets ($-6.0, -6.0, -5.1$ on the height-setting joints) cut that to 4.2%. **v11**, round 3 fine-tuned with the 87 calibrated block-stacking episodes at 9%, scores 472 vs 483 with every suite within noise: calibrated real data no longer costs the twin.
- **Best-of-8 consensus** (medoid of 8 sampled chunks, round 2): held_out_objects 55 vs 50 ($p = 0.41$), clean_many 87 vs 86; within noise at 8× inference.
- **DiVeR verifier** (best-of-8 on round 3, after arXiv 2610.04933): an MLP on pooled backbone features, joints and the chunk, trained on 300 round-3 rollouts (226 succeeded) with disagreement-weighted BCE, ranks held-out steps at AUC 0.79; picking with it scores held_out_objects 54 vs 57 ($p = 0.66$) and clean_many 85 vs 88 ($p = 0.51$). Scoring each candidate by the action expert's own state (AUC 0.77) is worse on held_out_objects (48, $p = 0.049$); adding 305 real ArmnetBench outcomes (87 successful) leaves it within noise (52, $p = 0.33$).
- **RL expert** (to replace the scripted expert): plain PPO on privileged state reached 1% held-out recovery by 4.4M steps and was stopped; the reward paid hovering more than stacking. A residual PPO policy on the scripted expert, with potential-based shaping, recovers 74.0% of held-out failure snapshots against the expert's 45.9%.

3 What is not established

- **Nothing has run on the physical arm.** Every number is from the MuJoCo twin.
- **The new-object gain is suggestive, not established.** Round 3's held_out_objects was 61 then 57 against round 2's 50 over two training runs; only the first is significant on its own ($p = 0.043$, uncorrected).
- **Round 2's gain is mostly one suite.** Only real_1_pomade_tin is significant on its own; the other five are better but within noise. The pooled gain passed a gate fixed in advance.
- **The FLUX failure mechanism is a hypothesis.** Conditioning on its own 8-frame history and integrating deltas fits the slow-start, no-contact pattern, but has not been tested by retraining.
- **so101-benchmark remains 0–1 of 50** for every policy; the external gap is open.
- **Single seed** for every benchmark (seed 42); no checkpoint was trained twice.

4 Next

Round 3 retrained and promoted (483); RL-expert recoveries in rounds 4 and 5 (×8, ×2) scored 471 and 455 and are dropped. Best-of-N selection (medoid, DiVeR verifier) is closed. Next: failure-state synthesis with the scripted expert; and an RL expert that can recover dropped and held blocks. Code, PRs and runs: https://github.com/charleneleong-ai/nebius-physical-ai, W&B project `chaleong/preflight`.